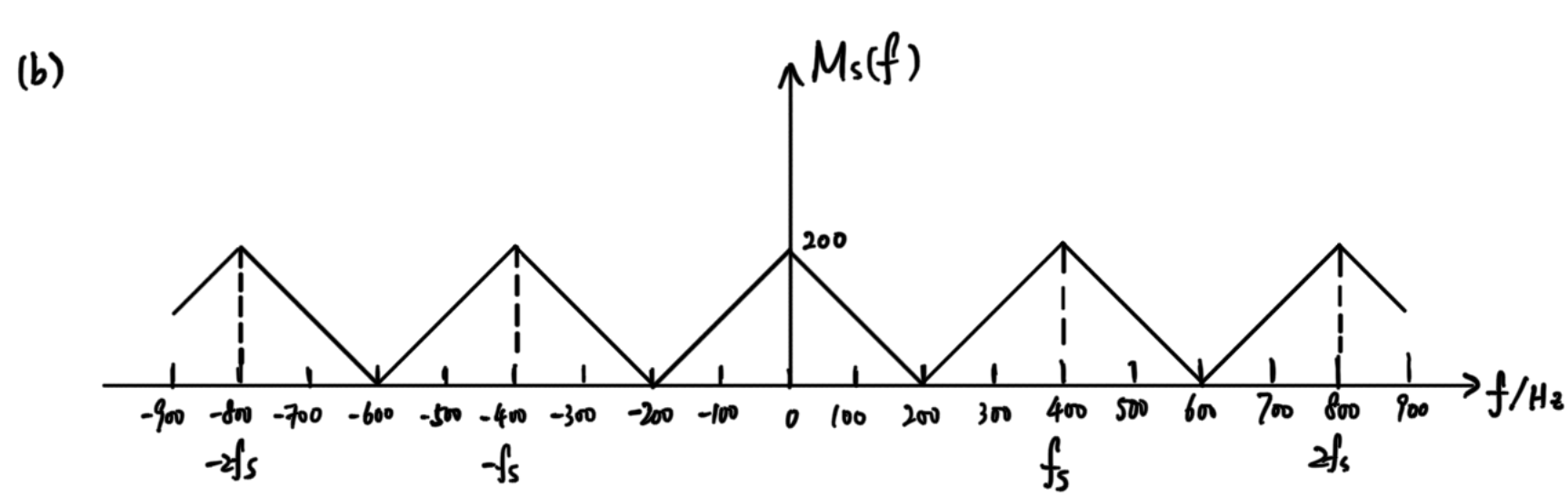
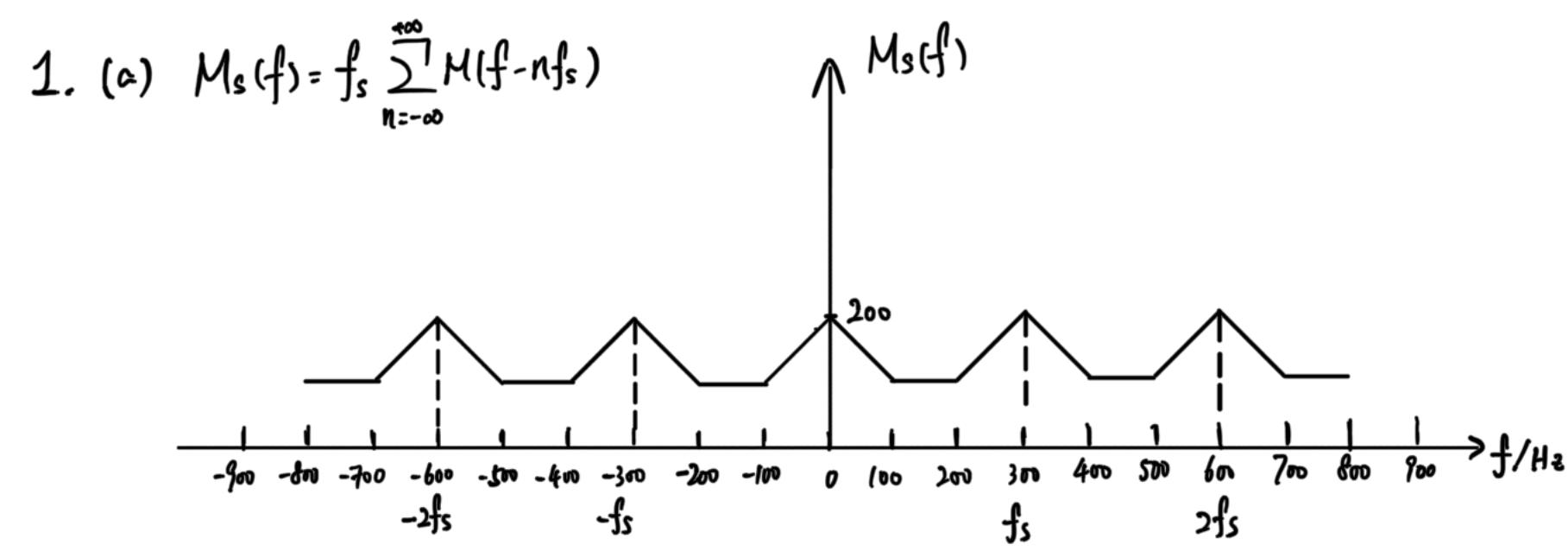
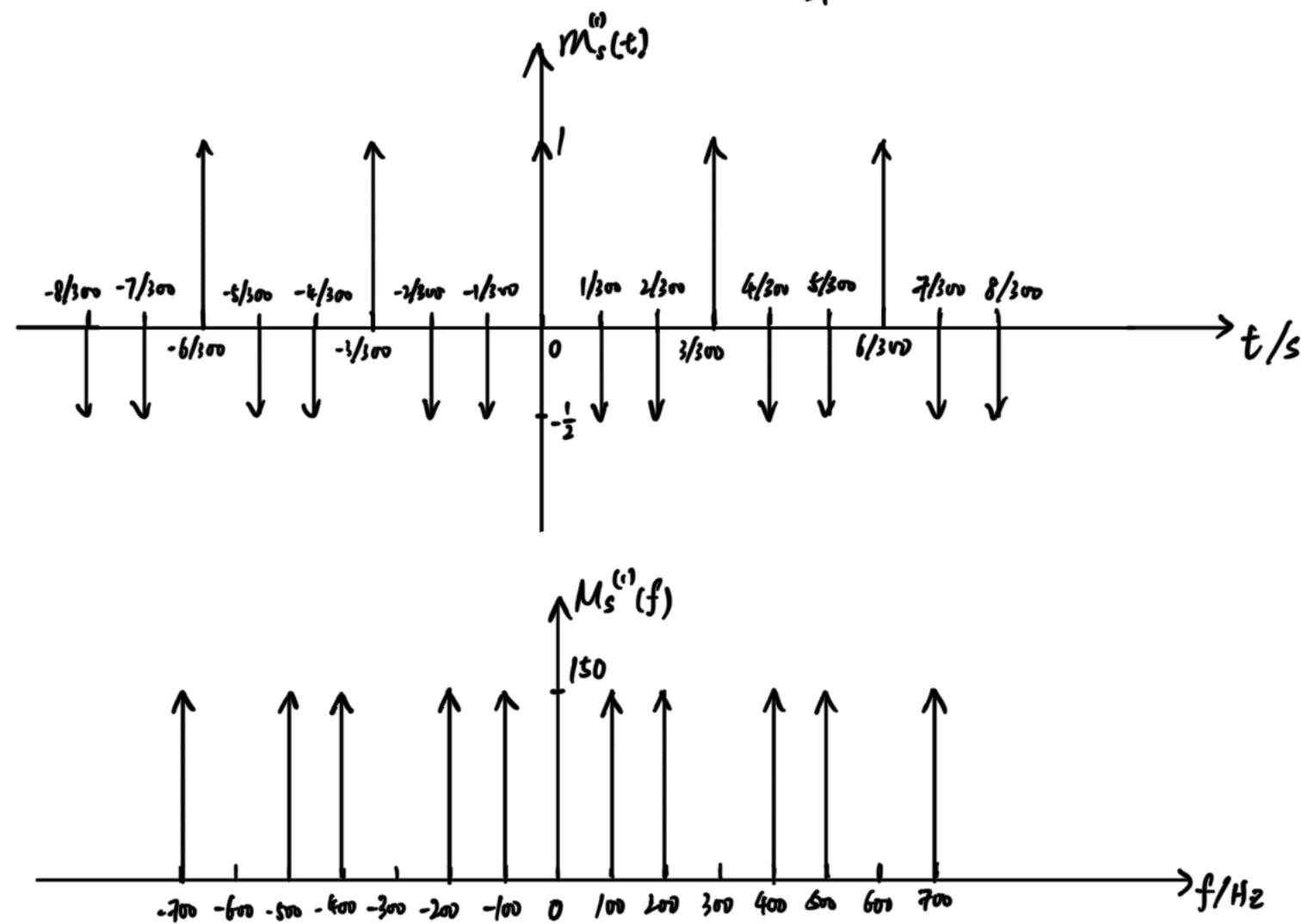
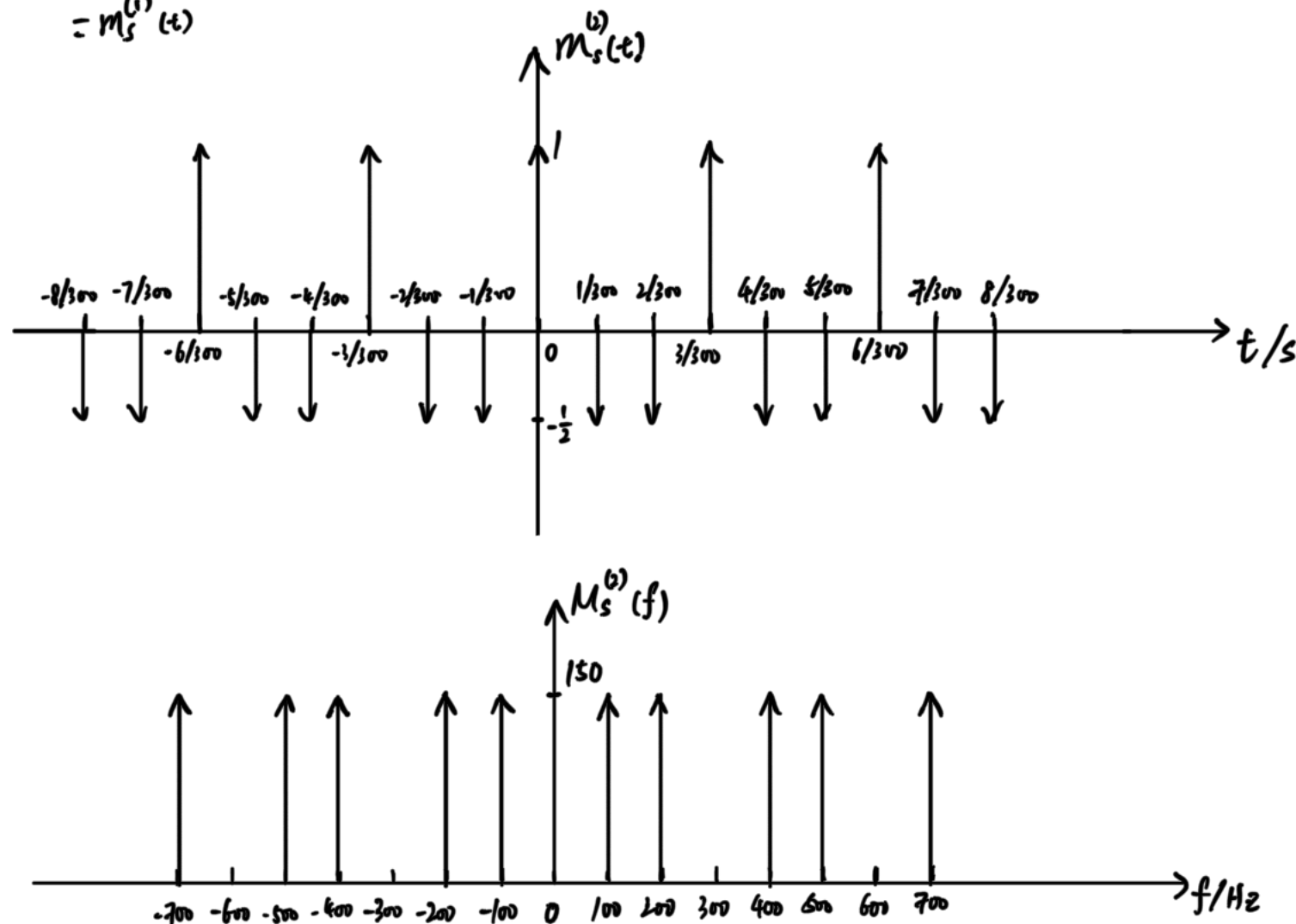




2. (a) $m^{(1)}(t) = \cos(200\pi t) = \cos(2\pi \cdot 100 \cdot t)$
 $m_s^{(1)}(t) = \cos(2\pi \cdot 100 \cdot t) \sum_{n=-\infty}^{\infty} \delta(t - \frac{n}{300}) = \sum_{n=-\infty}^{\infty} \cos(\frac{2\pi n}{3}) \delta(t - \frac{n}{300})$
 $M^{(1)}(f) = \frac{\delta(f+100) + \delta(f-100)}{2} \Rightarrow M_s^{(1)}(f) = f_s \sum_{n=-\infty}^{\infty} M(f - nf_s)$
 $= 150 \sum_{n=-\infty}^{\infty} (\delta(f+100-300n) + \delta(f-100-300n))$
 $= 150 \sum_{k=-\infty}^{\infty} \delta(f+100k)$



(b) $m^{(2)}(t) = \cos(2\pi \cdot 200 \cdot t)$
 $m_s^{(2)}(t) = \sum_{n=-\infty}^{\infty} \cos(\frac{4\pi n}{3}) \delta(t - \frac{n}{300})$
 $= \sum_{n=-\infty}^{\infty} \cos(\frac{2\pi n}{3}) \delta(t - \frac{n}{300})$
 $= m_s^{(1)}(t)$



(c) We can see that $m_s^{(1)}(t) = m_s^{(2)}(t)$
 According to the Sampling Theorem
 $2f_2 = 2 \times 200 \text{ Hz} = 400 \text{ Hz} > 300 \text{ Hz} = f_s > 2 \times 100 \text{ Hz} = 2f_1$
 i.e. $2f_2 > f_s > f_1$

Therefore, $m_s(t)$ can be reconstructed
 $m_s(t)$ will be reconstructed to $m^{(1)}(t)$

$\Rightarrow m_r(t) = m^{(1)}(t) = \cos(200\pi t)$

3. $SNR_g = \frac{\sigma_m^2}{\frac{N_{noise}}{3 \cdot 2^{2R}}} = 3 \times 2^{2R} \times D^2$
 $10 \log SNR_g = 4.77 + 20R \log 2 + 10 \log (\frac{1}{2})$
 $= 4.77 + (10 \log 2)(2R - 1)$
 $= 4.77 + 3.01(2R - 1) \geq 30 \text{ dB}$
 $R \geq \frac{1}{2} \left(\frac{0.5 \cdot 33}{3.01} + 1 \right)$
 $\Rightarrow R_{min} = \left\lceil \frac{1}{2} \left(\frac{0.5 \cdot 33}{3.01} + 1 \right) \right\rceil = 5$

4. $p(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

(a) $p(x)$ is symmetric along $x=\mu$

By intuitive, $D_2^{opt} = \mu$, $D_1^{opt} = -\infty$, $D_3^{opt} = +\infty$

$T_1 = \frac{\int_{-\infty}^{\mu} x p(x) dx}{\int_{-\infty}^{\mu} p(x) dx} = \frac{\mu \int_{-\infty}^{\mu} p(x) dx + \int_{-\infty}^{\mu} (x-\mu) p(x) dx}{1/2}$
 $= \mu + 2 \int_{-\infty}^0 x \cdot \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right) dx$
 $= \mu + 2 \int_{-\infty}^0 \frac{\sigma}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right) d\left(\frac{x^2}{2\sigma^2}\right)$
 $= \mu + \sqrt{\frac{2}{\pi}} \sigma \exp\left(-\frac{x^2}{2\sigma^2}\right) \Big|_0^{-\infty}$
 $= \mu - \sqrt{\frac{2}{\pi}} \sigma$
 $T_2^{opt} = \frac{\int_{\mu}^{+\infty} x p(x) dx}{\int_{\mu}^{+\infty} p(x) dx} = \frac{\mu \int_{\mu}^{+\infty} p(x) dx + \int_{\mu}^{+\infty} (x-\mu) p(x) dx}{1/2}$
 $= \mu + \sqrt{\frac{2}{\pi}} \sigma$

(b) $\sigma_g^2 = \int_{-\infty}^{\mu} (x-\mu + \sqrt{\frac{2}{\pi}} \sigma)^2 p(x) dx + \int_{\mu}^{+\infty} (x-\mu - \sqrt{\frac{2}{\pi}} \sigma)^2 p(x) dx$
 $= \int_{-\infty}^{\mu} (x-\mu)^2 p(x) dx + \frac{2}{\pi} \sigma^2 \left(\int_{-\infty}^{\mu} (x-\mu) p(x) dx - \int_{\mu}^{+\infty} (x-\mu) p(x) dx \right) + \frac{2}{\pi} \sigma^2 \int_{-\infty}^{+\infty} p(x) dx$
 $= (1 + \frac{2}{\pi}) \sigma^2 - \sqrt{\frac{2}{\pi}} \cdot 2\sigma \cdot \frac{\sigma}{\sqrt{2\pi}} \cdot 2$
 $= (1 + \frac{2}{\pi} - \frac{4}{\pi}) \sigma^2 = (1 - \frac{2}{\pi}) \sigma^2$

5. (a) $f_s \geq 2 \times f_{max} = 8 \text{ kHz}$

(b) $\frac{2f_H}{K} \leq f_s \leq \frac{2f_L}{K-1}$, $K=1, 2, \dots, \lfloor \frac{f_H}{B} \rfloor$, $B=1 \text{ kHz}$

$f_{s,min} = 2B \left(1 + \frac{\{f_H/B\}}{\lfloor f_H/B \rfloor} \right)$

For each sub-bands, $\{f_H/B\} = 0$

$\Rightarrow f_{s,min} = 2B = 2 \text{ kHz}$

(c) $-314 < 0 \Rightarrow M_1 = 0$

$256 < 314 < 512 \Rightarrow (M_2, M_3, M_4) = (1, 0, 0)$

$314 - 256 = 58$, $64 > 58$, $32 + 16 < 58 \Rightarrow (M_5, M_6, M_7, M_8) = (0, 0, 1, 1)$

PCM Code = 01000011

Representation point: $-(256 + 32 + 16 + 8) = -312$

error = $[-312 - (-314)] \Delta = 2\Delta$

noise = $4\Delta^2$

(d) bit-rate = $2 \text{ kHz} \times 8 \text{ bits} \times 4 \text{ channels}$
 $= 64 \text{ kbps}$